Chapter 3

Detrending Datasets

A given dataset is often the outcome of a deterministic part mixed with random fluctuations about the deterministic component. The physics of the deterministic part is often much easier to understand and may even obey known equations. In certain datasets, the impact of the deterministic factor is so dominant that the need for detrending is important to isolate the role of the fluctuating part. Understanding the random fluctuations, sometimes referred to as noise, can oftentimes be challenging. To investigate the memory properties that could be present in a dataset, the isolation of the random fluctuations from the deterministic contribution is often needed. Separating the deterministic part from the random fluctuations is often referred to as detrending. There are many ways of detrending a dataset but, as examples, we will discuss in this chapter four methods: (a) using best-fit curves; (b) employing moving averages; (c) looking at the power spectral density; and (d) applying a cycle averaging analysis which is also called fingerprinting. Note, however, that not all datasets need detrending and a judgment has to be made if it should be applied to certain physical systems depending on the aim of the investigation.

3.1 Best-fit Curves

Using best-fit curves to detrend a dataset is probably the easiest to understand. Given a plot of the dataset, we could superimpose a line or a curve that best captures the dataset. By subtracting from the raw values of the dataset the values of the best-fit curve, we generate fluctuating deviations of the dataset from the best-fit curve. These series of numbers deviating from the best-fit curve could be plotted and investigated for possible memory properties using the stochastic model discussed in Chapter 2.

Consider as an example the daily record of the global atmospheric carbon dioxide (CO_2) concentration measured at the Mauna Loa Observatory in Hawaii since 1958 (www.esrl.noaa.gov). The plot of the CO_2 concentration, known as the Keeling curve in honor of Charles David Keeling (1928-2005), shows oscillations and periodicity as a function of changes in seasons and differences between night and day. The weekly mean CO_2 data from May 1974 to August 2017, is shown as the red graph in Figure 3.1.

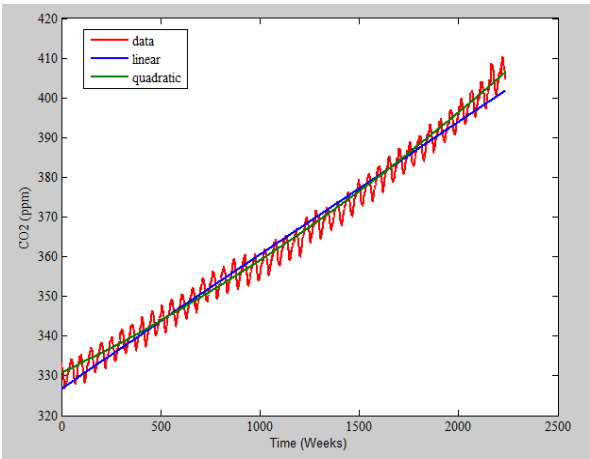

Figure 3.1. Weekly mean of CO_2 concentrations (red) from May 1974 to August 2017 with a linear fit (blue) and a quadratic fit (green).

One way of detrending is to put a linear fit (blue line), or alternatively a quadratic fit (green line), for the data in red. Each CO_2 data point can then be subtracted from the linear fit generating fluctuations as shown in Fig. 3.2 (blue color). For comparison, subtracting the datapoint values from the quadratic fit also gives rise to the fluctuations in Fig. 3.2 (green color). The mean square deviation of the fluctuations in Fig. 3.2 can then be evaluated as discussed in Chapter 4.

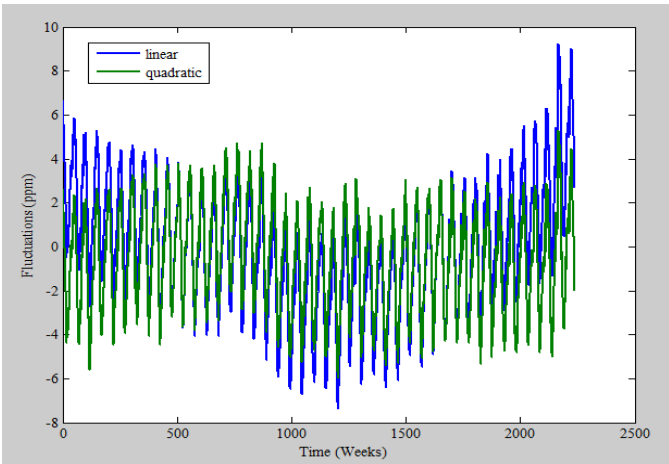

Figure 3.2. Fluctuations obtained by subtracting the CO_2 data points from the linear fit (blue) and quadratic fit (green).

3.2 Moving Averages

Detrending a dataset using moving averages is conceptually straightforward. Consider as an example the time series for the monthly mean for sunspot numbers from solar cycle 1 to solar cycle 24. Figure 3.3 shows the raw data as gray dots and the 7-month average as blue curves.

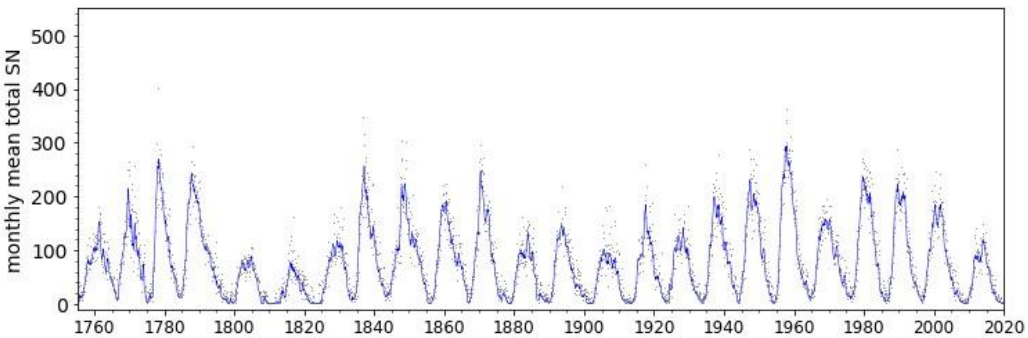

Figure 3.3. Monthly sunspot numbers (gray dots) and 7-month moving average (blue curves) from solar cycle 1 to solar cycle 24. Each solar cycle lasts 11 years on the average.

Subtracting the raw data from the moving average yields the fluctuations which becomes the working data for investigation. The fluctuations or residuals for a moving average of 7 months is shown in Fig. 3.4. The residuals have a mean of 5.87 and standard deviation of 20.75. The mean square deviation for the residuals is then plotted against time and compared with a theoretical MSD.

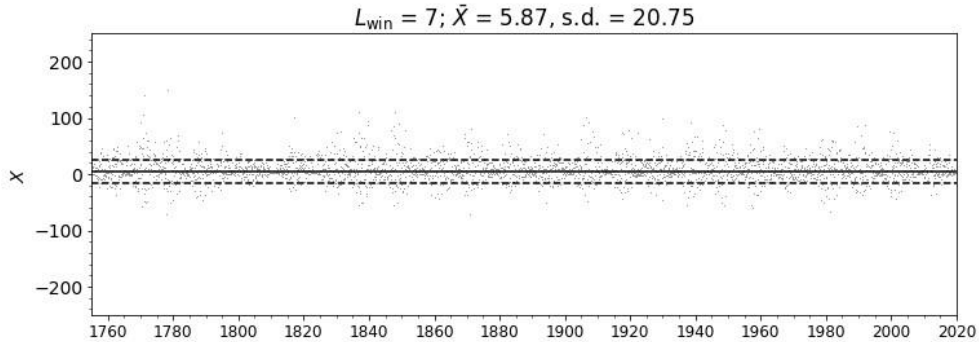

Figure 3.4 The residual time series for monthly sunspot numbers with a mean of 5.87 and standard deviation of 20.75.

3.3 Power Spectral Density Analysis

The deterministic component of a dataset sometimes dominate a signal and the power spectral density (PSD) analysis helps separate the deterministic part from the random fluctuations (Guler et al 2005, Peng et al 2014). For instance, given a time series, one can take its Fourier transform to obtain the frequency domain. Consider the time series for the population of algae in experiment C9 of the paper by Blasius et al (2020) on predator-prey systems. The PSD of the time series can be calculated by taking the normalized squared magnitude of the frequency domain (Fig. 3.5). From a plot of the PSD versus frequency, the peaks occur at dominant frequencies (periodic component) of the signal. To remove the dominant or deterministic part, a PSD threshold can be specified so that we can identify the frequencies of the peaks that go beyond the threshold. In Fig. 3.5, we illustrate four possible thresholds at 1.0, 0.2, 0.3 and 0.5. We then set to zero the frequencies corresponding to the peaks above the chosen threshold. By applying an inverse Fourier transform, we reconvert the filtered signal to the time domain where only the desired fluctuations remain (see, Fig. 3.6, bottom).

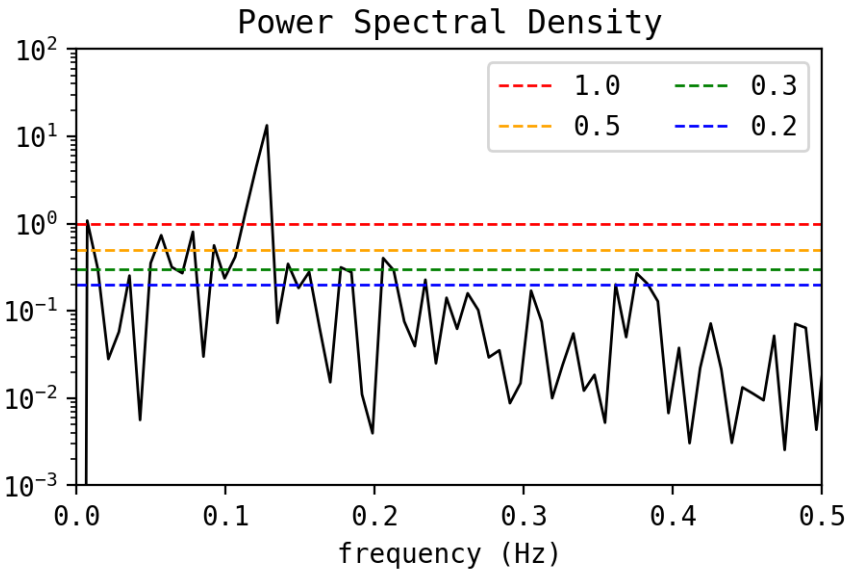

Figure 3.5. Applying a Fourier transform to a time series gives the power spectral density versus frequency plot. The dominant prominent peaks can be removed by choosing a cut-off value. Different possible cut-offs are: 1.0, 0.2, 0.3, or 0.5.

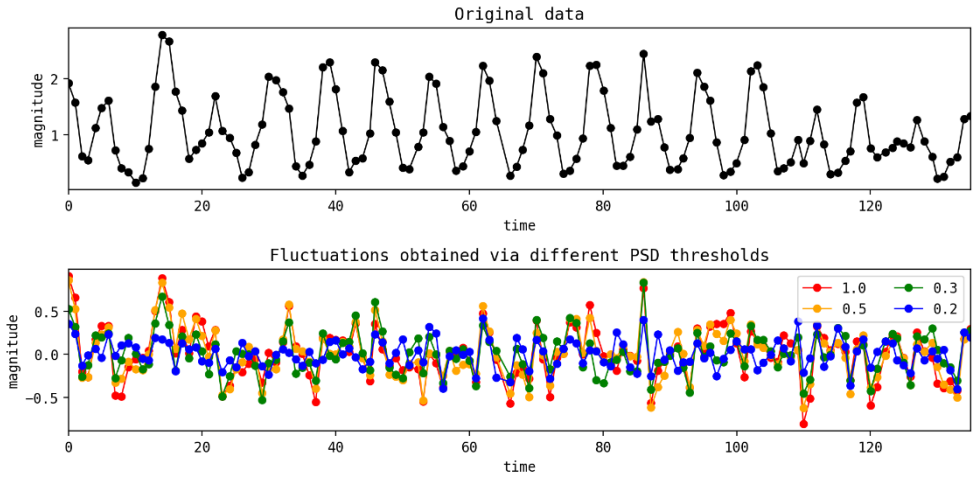

Figure 3.6 The original fluctuating dataset (top). The dominant deterministic part is removed using the power spectral density analysis leaving behind a seemingly random fluctuation as remnants (bottom). The colors of the fluctuations correspond to the different possible cut-offs (1.0, 0.2, 0.3, or 0.5, see, Fig. 3.5) which eliminate the frequency of the dominant or deterministic component.

3.4 Cycle Averaging Analysis: Fingerprinting

Another method of detrending is the cycle averaging analysis which is also referred to as fingerprinting a repeating cycle of a dataset (Jutras and Howard 2015, Cole et al 2009). The average of the repeated cycle is then subtracted from the raw data leaving only the fluctuations about the average. The fluctuations can then be investigated for possible memory inherent in the system.

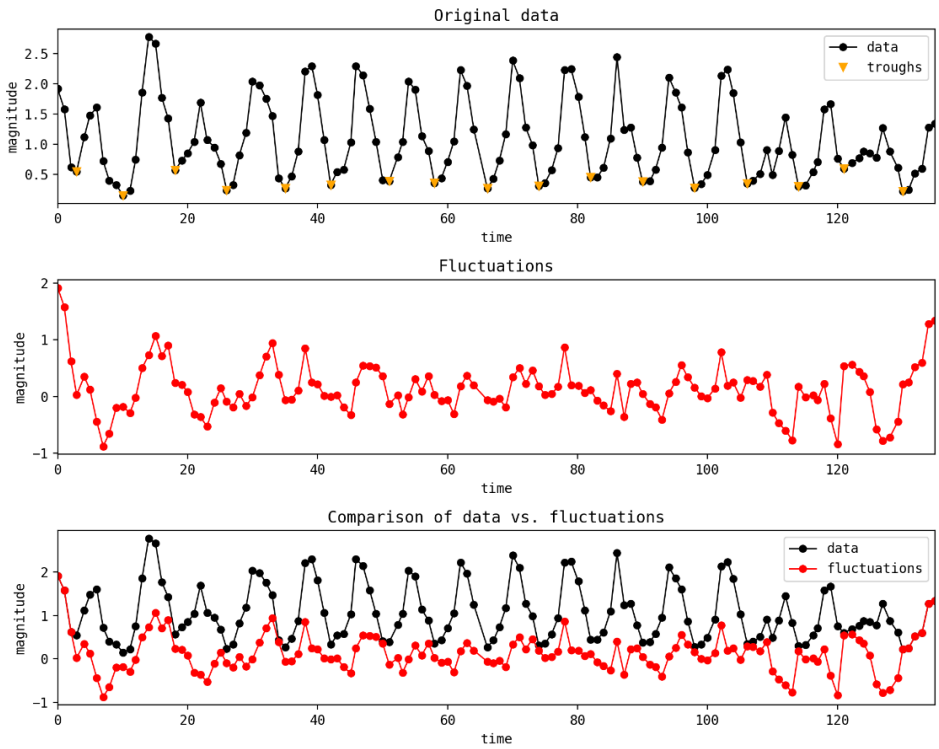

Figure 3.7. The original fluctuating dataset exhibiting a dominant cycle (top). The remaining fluctuation after the dominant deterministic part is removed (middle). Comparison of the original data and the remaining fluctuations (bottom).

Consider a fluctuating cyclic signal shown in Figure 3.7 for the algae population in experiment C9 of a predator-prey system (Blasius et al 2020). We can identify and separate each cycle by cutting them at the trough indicated by the yellow dots in Fig. 3.7 (top). To apply a cycle averaging, we collect all the cycles, or segments, by aligning their maxima. The alignment of maxima may require padding or adding zeroes at the start of some segments. The lengths of all segments are then made equal by also adding or padding zeroes at the end of the cycle (Fig. 3.8, middle). The fingerprint of all the cycles would be the segment mean calculated as the average of all segments equalized in length with aligned maxima (Fig. 3.8, right). The fluctuations of the signal are then obtained by subtracting the mean or fingerprint from each of the original segments (Figure 3.7, middle and bottom). Note, however, that before subtracting the mean, the paddings from the segments are removed, as well as the values of the segment mean at the padding indices, so no new data points are added to the fluctuations. The next step is to get the mean square deviation of the fluctuations as discussed in Chapter 4.

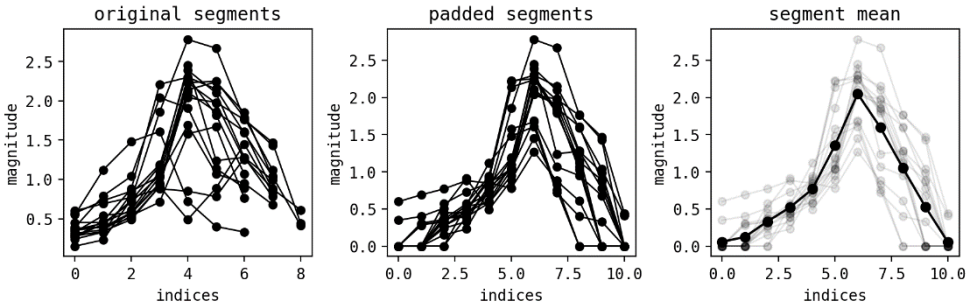

Figure 3.8. The original segments or cycles are collected in preparation for averaging (left). The maxima of the segments are aligned by padding or adding zeroes at the start. The length of all segments is also equalized by padding the end of the segments (middle). The cycle average or fingerprint (right).